Basics of statistical physics: Micro- and macrostates, entropy, ensembles

May 24, 2026

Boxed text corresponds to material that goes beyond what was covered in the lecture.

1 Phase space, micro-and macrostates

Let us consider a system made out of N particles enclosed in volume V. We will treat them as classical point particles, obeying Newton's laws of motion. As such, to fully specify the state of the system at a given time, we need to fix the positions $\vec{x}_i$ and velocities $\vec{v}_i$ of all the particles, $i = 1, \ldots, N$. Specifying these is sufficient the determine the state of the system at any later time via Newton's equations, assuming we know the interactions between the particles. Instead of the velocities, it is often convenient to specify the corresponding *momenta*, $p_i := m_i \vec{v}_i$ where m_i is the mass of the i-th particle. The positions $\vec{x}_i$ and momenta p_i together constitute the the system's *phase space*. This is a $6N$ dimensional space (since each vector has 3 components). Each point of the phase space is a possible state of the system and following the state as a function of time corresponds to a trajectory in phase space.

Treating the system as isolated, their total energy will be conserved over time and we will equate this with the *internal energy* U appearing in the thermodynamical description. Assuming (as is generally the case) only 2-body interactions between particles we can write the total energy as

$$U = E(\vec{x}_1, \vec{p}_1, \ldots \vec{x}_N, \vec{p}_N) = \underbrace{\sum_{i=1}^{N} \frac{|\vec{p}_i|^2}{2m_i}}_{E_{\text{kin}}} + \underbrace{\frac{1}{2} \sum_{i,j=1}^{N} u(\vec{x}_i - \vec{x}_j)}_{E_{\text{int}}}, \tag{1.1}$$

where E_{kin} is the total kinetic energy and E_{int} is the energy from the interparticle interactions, and we assumed that the potential energy (corresponding to external gravitational, electromagnetic etc. fields) can be ignored, in line with the idea that the system is well isolated from its environment[1]. We will use E for the energy when we consider it as a function over phase space (i.e., a function of the positions and momenta) and reserve U for the internal energy considered as a function of the system's macrostate (to be defined below).

In principle, the position and momentum coordinates can vary continuously. However, in any practical situation, we can only resolve them within some finite resolution. Let us denote by Δx the smallest differences in position we can resolve and Δp the smallest differences in momenta. This suggests that we can divide the phase space of all possible states into elementary *cells* with "volume" $(\Delta x)^3(\Delta p)^3$: states in distinct cells can be distinguished while states within the same cell cannot. Quantum mechanics puts a fundamental limitation on measurement precision of the form $\Delta x \Delta p \geq \hbar/2$; one therefore often takes the volume of the phase space cells to be (ignoring constant prefactors) $(\Delta x)^3(\Delta p)^3 = h^3$. In the following, we will refer to the phase space cell corresponding to the state of the system as the *microstate*. While, due fo the finite resolutions Δx and Δp, knowing the microstate does not fully specify all the variables, it still involves knowledge of all the individual particles.

[1] For electric fields, this can be achieved with a Faraday cage. For gravitational fields, such screening might not be possible, but in many cases gravitational effects can nevertheless be ignored.

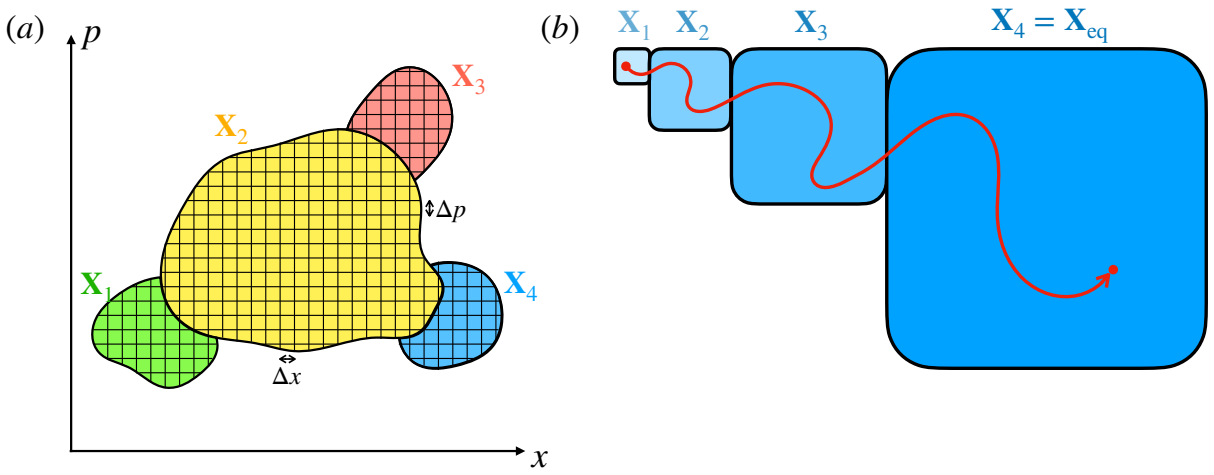


Figure 1: Micro-and macrostates in a system of N particles. (a) Illustrates the division of phase space: small boxes correspond to phase space cells of size $(\Delta x \Delta p)^{3N}$; these consistute the system's microstates. Large colored regions are the different macrostates, each containing (exponentially) many microstates. (b) Increase of entropy en route to reaching thermal equilibrium. The system's trajectory (red line) passes to larger and larger macrosates, coresponding to an increase of entropy, until the equilibrium state with maximal entropy is reached.

For a system with a fixed number of particles N in a fixed volume V with a fixed energy U, the total number of possible microstates is finite[2] However, this number will grow exponentially with N[3]. Thus, in the following we will take as our starting point a physical system with a finite but large number of possible microstates and an energy E that is a function of the microstate[4].

For a macroscopic system, with $N \gg 10^{20}$ particles, knowing the system's microstate is clearly impossible in practice. It is also unnecessary: we only really care about a few macroscopic properties that are experimentally accessible. What precisely those macroscopic observables are depends on the particular physical situation, which determines what we can and want to measure. The total particle number N and energy U are examples of such macroscopic quantities. Others might include, for example, more detailed information about the spatial distribution of these quantities, such as the number of particles and the amount of energy contained in different parts of the system. The total number of these macrostates is smaller than those of the microstates; in particular we will assume that the macrostates are defined in such a way that their total number is independent of the number of particles.

> While the discussion below will not depend on the details of how the macrostates are defined, we can take the following example as representative. Consider dividing the system into M different parts, labeled by an integer $m = 1, \ldots, M$. We will assume that the volume of each part (denoted by V_m) contains some finite fraction of the total volume V. Let N_m, U_m denote the number of particles and the energy in part m; the corresponding *densitites* are given by $\rho_m = N_m/V_m$ and $\epsilon_m = U_m/V_m$. We can imagine that these densities can be measured with some precision $\Delta\rho$ and $\Delta\epsilon$. We can then specify a macrostate by specifying the densities in each part within the given precision. Assuming that $M, \Delta\rho, \Delta\epsilon$ are all taken to be independent of N, then the number of macrostates will also remain finite in the limit $N \to \infty$.

We thus imagine that the system's phase space is divided into some finite number of macrostates. Each of those will contain some large number of microstates (phase phase cells): indeed, given the large difference between the number of micro- and macrostates, most macrostates will contain exponentially

[2]Note that fixing V and U limits the range of possible values of positions and momenta.

[3]To see this, consider particles in a box of size $L \times L \times L$. The x-coordinate of any given particle can take $L/\Delta x$ different values within measurement precision Δx. But in order to specify the phase space cell, we have to specify the x-coordinate of all the particles separately, for which there are $(L/\Delta x)^N$ choices.

[4]This description also generalizes to other physical systems, where the microstates do not originate from the phase space cells defined above. An example are "classical spins" used in simplified models of magnetism.

many in N cells, made up by all the different points in phase space that are consistent with a given macrostate. For a given macrostate $\mathbf{X}$ we will denote the number of corresponding microstates by $\Omega(\mathbf{X})$. We can think of $\Omega(\mathbf{X})$ as the phase space volume associated with $\mathbf{X}$, measured in units of the volume of the phase space cells. This division of the phase space is illustrated in Fig. 1(a).

2 Equilibration, Boltzmann entropy and second law

So far, we have divided the set of all microstates (associated to phase space cells, of which there are exponentially many) to a much smaller number of macrostates. As the system evolves in time, according to Newton's laws, its microstate will change due to the particles moving around and exchanging momenta due to their interactions (and, potentially, due to interactions with the walls of the container separating the system from its environment). During this evolution, the macrostate might also change, if the system's trajectory crosses the boundary separating different macrostates in phase space. In particular, our expectations from thermodynamics suggest that, given sufficient amount of time, the macrostate should approach a particular *equilibrium* state[5]. What property distinguishes this equilibrium state from all the other macrostates? This question is answered by the following fundamental assumption:

Fundamental assumption (Boltzmann version): The equilibrium macrostate $\mathbf{X}_{\rm eq}$ is the one with the largest volume: $\Omega(\mathbf{X}) = \max.$ when $\mathbf{X} = \mathbf{X}_{\rm eq}$.

Why should such a general assumption hold? Among other things, why should the macrostate settle down to one particular value, rather than keep changing forever as the system's trajectory explores its phase space? The answer lies in the particular structure of macrostates. While the volume Ω is exponentially large in N for almost all macrostates, these volumes tend to be vastly different for different macrostates. Consider a simple example where the system is divided into two equally sized parts (referred to as "left" and "right") and the macrostate $\mathbf{X}$ is specified by the fraction of particles $n_L = N_L/N$ on the left. Consider two different macrostates, corresponding to two different values of this fraction, denoted by n_L and n'_L. Ignoring the interactions between particles[6], the phase space volume is determined by the number of ways we can choose the N_L particles on the left, $\Omega(n_L) \propto \binom{N}{N_L}$. An application of Stirling's approximation shows that for large N this can be well approximated by a function of the form $\binom{N}{n_L N} \approx e^{N h(n_L)}$ where $h(n_L)$ is a monotonically growing function in the range $n_L \in (0, 1/2]$. In other words: as we increase n_L towards $1/2$, the number of available microstates *grows exponentially*. In particular, the vast majority of microstates correspond to the macrostate $n_L = 1/2$, the equilibrium state[7].

This paints the following picture of equilibration: on a microscopic level, the system explores its phase space, wondering from one cell to another during its trajectory. If the initial state is away from thermal equilibrium (e.g. $n_L^{\rm init} < 1/2$ in the example above), then it will correspond to a tiny fraction of all possible microstates and the system will soon cross over to a different macrostate with an exponentially larger volume (e.g. a state with $n_L > n_L^{\rm init}$). This happens simply because this latter macrostate is much larger; it would require a special explanation for the system *not* to transition to a different macrostate. This process will continue, with the system going from one macrostate to another, corresponding to increasingly large phase space volumes, until it finally reaches the equilibrium state with the largest volume. Once it is there, since this equilibrium state contains almost all of the microstates, the system will stay there, wondering around the different microstate corresponding to the same equilibrium macrostate essentially forever. The word "essentially" is important here: given enough time, the system will accidentally get out of equilibrium and into a different macrostate. However, this will typically take a time that is exponentially long in N and for a macroscopic system, composed of $N \gg 10^{20}$ particles, this will take far longer than the age of the universe. This picture of equilibration is illustrated in Fig. 1(b).

Comparing the above fundamental assumption with the known laws of thermodynamics, we can arrive at a microscopic interpretation of entropy. From thermodynamics, we know that, in a closed system, entropy increases monontonically in time, reaching its maximum value in equilibrium. The above argument shows that something similar happens with the phase space volume Ω, suggesting that

[5] E.g., in the example given above, the distribution of particles and energy should become homogeneous, with the densities ρ_m and ϵ_m all approaching the same value independent of m.

[6] Taking the interactions into account complicates the calculation, but it does not change the overall conclusion.

[7] More precisely, consider all microstates where $|n_L - N/2| > \delta$ for some fixed constant δ. The fraction of such states amongst all microstates goes to zero exponentially in the limit $N \to \infty$.

we should interpret the entropy of a macrostate in terms of its phase space volume. As the above considerations also show, Ω tends to grow exponentially with N, while we expect that the entropy is in general extensive, growing linearly with N. This suggests the following identification between the two, known as the *Boltzmann entropy formula*:

$$S_{\text{Boltzmann}}(\mathbf{X}) := k_{\text{B}} \ln \Omega(\mathbf{X}). \tag{2.1}$$

Here, we introduced *Boltzmann's constant* $k_B = 1.381 \times 10^{-23} \frac{\text{J}}{\text{K}}$. It makes the units of the entropy match those from macroscopic thermodynamics. In some sense it is needed only for historical reasons: historically, energy and temperature were defined independently, and their close relationship was only defined later, with the advent of statistical physics. Its small numerical value reflects the fact that macroscopic objects contain a large number of particles, comparable to Avogadro's number.

Comparing the Boltzmann entropy definition (2.1), with the above picture of how the system reaches thermal equilibrium, provides a microscopic explanation for the second law of thermodynamics: the system's entropy tends to increase over time, since it naturally goes from smaller to larger macrostates as it evolves.

Equation (2.1) provides a point of contact between statistical physics and thermodynamics. With this being established, we can define other thermodynamic quantities in terms of the microscopic description. In equilibrium, the entropy is specified by the total volume V, the number if particles N and the internal energy U[8]. To be consistent with thermodynamics, we should then define the temperature T, the pressure P and the chemical potential μ in terms of the three partial derivatives of S as follows:

$$\frac{1}{T} := \frac{\partial S}{\partial U}, \qquad P := T\frac{\partial S}{\partial V}, \qquad \mu := T\frac{\partial S}{\partial N}. \tag{2.2}$$

This, together with Eq. (2.1), gives a microscopic interpretation of the various quantities appearing in thermodynamics, from which further quantities (compressibility, heat capacity, etc.) can be derived.

3 Gibbs approach, microcanonical distritubion

So far, we have been following what might be called the "Boltzmann approach" to statistical physics. In this approach, we partition the microscopic phase space into large region corresponding to different macrostates, which correspond to the states appearing in the usual thermodynamic description. We imagine the microstate following a specific, well-defined trajectory which leads to a corresponding change in the macrostates.

An alternative posibility, which is also important in statistical physics is the "Gibbs approach". The idea behind it is that in any practical situation, determining the microstate of the system—even at the level of a particular phase space cell—is impossible. Instead, we only have imperfect knowledge of the system's state, based on a few macroscopic measurements. Therefore, we should describe the state *probabilistically*, by assigning different probabilities to the different microstates in a way that is consistent with the limited knowledge we obtained from our measurements.

In particular, let us assume that from our measurements we have determined the macrostate of the system to be a state $\mathbf{X}$ but we have no further knowledge of its microstates. In that case, it is natural to assign the same probability to all microstates consistent with $\mathbf{X}$ while assigning zero probability to other microstates. Using ω to label the system's different microstates, that is, the different phase space cells, this can be written as

$$p_\omega = \begin{cases} 0 & \text{if } \omega \notin \mathbf{X} \\ \frac{1}{\Omega(\mathbf{X})} & \text{if } \omega \in \mathbf{X} \end{cases} \tag{3.1}$$

We are primarily interested in the case where $\mathbf{X}$ is the thermal equilibrium state. As we observed above, this equilibrium state contains almost all of the available microstates. Therefore, up to an error that is exponentially small in N, we can replace the above distirubitoon with the following, so-called

[8] Note that what distinguishes V, N, U among physical quantities is that in the situation we are considering, for a closed system in a rigid container, they are all conserved, so that they have the same values for all microstates that the system visits during its evolution.

microcanonical distribution that depends only on the total internal energy U (for a fixed value of V and N):

$$p_\omega^{\mathrm{mc}} = \begin{cases} 0 & \text{if } E_\omega \neq U \\ \frac{1}{\Omega(U)} & \text{if } E_\omega = U \end{cases} \tag{3.2}$$

Here $\Omega(U)$ is the number of *all* phase space cells corresponding to states with energy U. In the microcanonical ensemble therefore all the microstates with the same energy U are equally probable. In the Gibbs approach, this is usually taken to be a fundamental assumption, on par with the fundamental assumption of the Boltzmann approach stated above:

Fundamental assumption (Gibbs version): A closed system in equilibrium is described by the microcanonical ensemble (3.2), in which all available microstates are equally probable.

While this distribution also assigns non-zero probability to macrostates that are not in equilibrium, their probability, by the previous considerations, is exponentially small in N; thus, the microcanonical ensemble correctly captures the system's properties in equilibrium up to negligibly small errors.

3.1 Gibbs entropy

We also need to adapt the definition of entropy to the Gibbs approach: in particular, instead of assigning entropy to a particular macrostate, we want to assign it to the probability distribution p_ω. Luckily, there exists such a definition, which was originally developed by Gibbs in his study of statistical physics. The same notion of entropy was later rediscovered by Shannon as a fundamental concept that characterizes probability distributions in the context of information theory. It is defined as follows:

$$S_{\mathrm{Gibbs}}[\{p_\omega\}] := -k_B \sum_\omega p_\omega \ln p_\omega. \tag{3.3}$$

What this notion of entropy does is that it quantifies the amount of uncertainty in a given probability distribution. When there is no uncertainty, i.e. we know the state of the system fully, then one of the p_ω is 1 and all other are 0 and the entropy is zero[9]. On the other hand, the entropy is maximal in the case of maximal uncertainty, when all p_ω are equal.

This last observation shows why we should indeed equate the Gibbs entropy with the usual thermodynamic entropy. If we consider a closed system, where the internal energy U is fixed, then in the microcanonical ensemble (3.2), which assigns equal probabilities to all microstates, the entropy will be maximal, as it should be in equilibrium. Conversely: if we accept Eq. (3.3) as the definition of entropy, it justifies the assumption of why the microcanonical distribution is the correct description of a closed system in thermal equilibrium: this is the distribution that maximizes the entropy. Indeed, in this case we find that the Gibbs definition of entropy coincides with the previous Boltzmann definition:

$$S_{\mathrm{Gibbs}}[\{p_\omega^{\mathrm{mc}}\}] = -k_B \sum_{\omega: E_\omega = U} \frac{1}{\Omega(U)} \ln \frac{1}{\Omega(U)} = k_B \ln \Omega(U). \tag{3.4}$$

4 Canonical distribution and Gibbs entropy

So far, we have considered closed systems that are completely isolated from their environments, such that no heat transfer occurs and no work is done on (or by) the system, making its internal energy U constant. Let us now turn to another (experimentally often more relevant) situation, when the system is in contact with a large external reservoir (or "heat bath") with which it can exchange heat. The reservoir is characterized by its temperature T_R (defined as in Eq. (2.2)). We will imagine that the reservoir is much larger than the system, so that its temperature does not change significantly due to its exchange of heat with the system. This situation is illustrated in Fig. 2

The combination of the system and the reservoir taken together constitute a closed system, to which our previous considerations apply. Its internal energy is $U_{\mathrm{tot}} = U + U_{\mathrm{R}}$ where U is system's internal energy and U_{R} is that of the reservoir. Let us fix a particular microstate ω of the system with energy E_ω. Then

[9] Note that while $\ln x$ is ill-defined when $x = 0$, the quantity $x \ln x$ has a well-defined limit, approaching 0 as $x \to 0$.

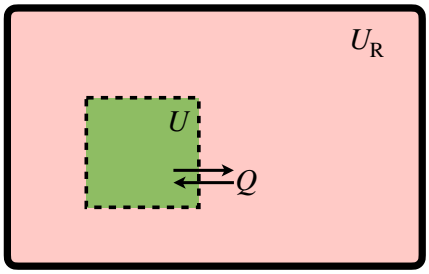


Figure 2: Setup for deriving the canonical distribution. The system (shown in green) is embedded into a large thermal reservoir (red) with which is can exchange heat so that the two maintain the same temperature. The system and reservoir together constitute an isolated system.

the total number of possible microstates of the reservoir is given by $\Omega_\mathrm{R}(U_\mathrm{tot} - E_\omega)$. The total number of microstates with energy U_tot is $\Omega_\mathrm{tot}(U_\mathrm{tot}) = \sum_\omega \Omega_\mathrm{R}(U_\mathrm{tot} - E_\omega)$. By our fundamental assumption, in equilibrium all microstates of the combined system are equally likely, so that the probability of observing a particular microstate ω of the system is given by the proportion of the number of states consistent with it. That is, the probability is given by

$$p_\omega = \frac{\Omega_R(U_\mathrm{tot} - E_\omega)}{\Omega_\mathrm{tot}(U_\mathrm{tot})} \tag{4.1}$$

We can now use the definition (2.1) for the entropy of the reservoir to write $\Omega_R(U_\mathrm{tot} - E_\omega) = e^{S_\mathrm{R}(U_\mathrm{tot} - E_\omega)/k_\mathrm{B}}$. By assumption, the system is much smaller than the reservoir, so we can take $E_\omega \ll U_\mathrm{tot}$ and expand the entropy to first order in E_ω. Using Eq. (2.2), the derivative is related to the temperature so we find $\Omega_R(U_\mathrm{tot} - E_\omega) \approx e^{S_R(U_\mathrm{tot})/k_\mathrm{B}} e^{-E_\omega/k_\mathrm{B}T_R}$. Here T_R is the temperature of the reservoir; however, in equilibrium, this must be the same as the temperature of the system so we will replace it with T henceforth. In the ratio (4.1), the first part, depending only on the properties of the reservoir, drops out and we are left with the *canonical distribution* over the microstates of the system:

$$p_\omega^\mathrm{can} = \frac{e^{-\frac{E_\omega}{k_\mathrm{B}T}}}{\sum_\omega e^{-\frac{E_\omega}{k_\mathrm{B}T}}} = \frac{e^{-\frac{E_\omega}{k_\mathrm{B}T}}}{Z}, \tag{4.2}$$

where we introduced the notation $Z = \sum_\omega e^{-\frac{E_\omega}{k_\mathrm{B}T}}$ for the normalization factor, the so-called *partition function*. Eq. (4.2) describes the correct probabilities to assign to a system that is in equilibrium with a large heat bath, or, equivalently, to a small part of a larger system in equilibrium.

Unlike the microcanonical distribution for a closed system, in the canonical distribution of Eq. (4.2), the total energy is not fixed. Instead, all possible values of the energy are possible, but they occur with different probabilities; indeed, what Eq. (4.2) reveals is that microstates with larger energy are exponentially less likely to appear. However, this exponential decay in the probability of individual microstates is compensated by the fact that in most systems, the number of possible microstates grows exponentially with the system's energy[10]. The competition between these two factors is controlled by the temperature T: when T is small, the probability of high-energy configurations is very strongly suppressed and the distribution is concentrated around low-energy configurations. As T gets larger, this suppression becomes weaker and weaker and the distribution is increasingly dominated by more high-energy configurations.

In fact, there is another way to motivate the canonical distribution (4.2), using the notion of Gibbs entropy defined in Eq. (3.3). It can be shown that if we consider all possible probability distributions that

[10]The reasons for this can be understood similar to the appearance of other exponentials we have encountered above: it originates from combinatorial factors having to do with the different ways in which we can distribute the same total amount of energy between different degrees of freedom.

have the same average energy $\overline{E} = \sum_\omega p_\omega E_\omega$, then the one that maximized the entropy is the distribution $p_\omega \propto e^{-\beta E_\omega}$ (ignoring an overall normalization factor). Here β is a constant that is set by $\overline{E}$; comparing with Eq. (4.2), we recognize it as the inverse of the temperature, $\beta = \frac{1}{k_\mathrm{B} T}$.

4.1 Energy and free energy

The canonican ensemble (4.2) gives a distribution over all the possible microstates ω of the system. It is worthwhile to also consider the distribution of total energy U. To To get this, the probability of individual microstates at a given energy has to be multiplied by the total number of such states:

$$p(U) = \sum_{\omega: E_\omega = U} p_\omega = \frac{1}{Z}\Omega(U) e^{-\frac{U}{k_B T}} = \frac{1}{Z} e^{-\frac{U - T S_B}{k_B T}}, \tag{4.3}$$

where S_B stands for the Boltzmann entropy, defined via Eq. (2.1). We recognize the combination $U - T S_B$ as form of (Helmholtz) *free energy*, familiar from thermodynamics. Eq. (4.3) says that the probability of observing a particular amount of total internal energy U in the system is set by a competition between energy and entropy expressed via this free energy.

In Eq. (4.3), the exponential decay of probabilities of *individual* microstates, encompassed in Eq. (4.2), is compensated by the exponential growth of available microstates as the energy is increased. As a result, the distribution $p(U)$ is peaked around some average energy $\overline{U}$ that is set by the temperature T. This average energy has a simple expression in terms of the partition function $Z(\beta) = \sum_\omega e^{-\beta E_\omega}$:

$$\overline{U} = \int \mathrm{d}U p(U) U = \sum_\omega p_\omega^{\mathrm{can}} E_\omega = \frac{1}{Z}\sum_\omega e^{-\beta E_\omega} E_\omega = -\frac{1}{Z}\frac{\partial Z}{\partial \beta} = -\frac{\partial \ln Z}{\partial \beta} \tag{4.4}$$

where we used that $\frac{\partial Z}{\partial T} = \sum_\omega e^{-\frac{E_\omega}{k_\mathrm{B} T}} \frac{E_\omega}{k_\mathrm{B} T^2}$.

How well does this average energy characterize the state of the system? To answer this question, we need to consider the size of the fluctuations around the average. These are quantified by the energy variance:

$$(\delta U)^2 \equiv \overline{U^2} - \overline{U}^2 = \sum_\omega p_\omega E_\omega^2 - \overline{U}^2 = \frac{\partial^2 \ln Z}{\partial \beta^2}. \tag{4.5}$$

We thus find that the quantity $\ln Z$ plays an important role, controlling various moments of the energy distribution through its derivatives (a property that continues to hold also for higher derivatives). What is the physical meaning of this quantity? To answer this, we need to evaluate the Gibbs entropy for the canonical distribution, which yields

$$S_{\mathrm{Gibbs}}^{\mathrm{can}} = -\frac{k_\mathrm{B}}{Z}\sum_\omega e^{-\frac{E_\omega}{k_\mathrm{B} T}} \ln\left(e^{-\frac{E_\omega}{k_\mathrm{B} T}}/Z\right) = \underbrace{\frac{1}{T}\frac{1}{Z}\sum_\omega e^{-\frac{E_\omega}{k_\mathrm{B} T}} E_\omega}_{=\overline{U}} + k_\mathrm{B} \ln Z. \tag{4.6}$$

Now, equating $\overline{U}$ with the thermodynamic internal energy, we can express the *Helmholtz free energy* F:

$$F = U - TS = -k_\mathrm{B} T \ln Z, \tag{4.7}$$

or, equivalently, $Z = e^{-\frac{F}{k_\mathrm{B} T}}$. Eq. (4.7) is a fundamental equation that expresses the relationship between macroscopic thermodynamics statistical physics in the canonical ensemble. From it, we see that the partition function Z, which originally appeared as a simple normalization factor, needed to ensure that probabilities sum up to 1, is in fact closely related to the free energy, which is the thermodynamic potential appropriate to the situation when the system is in thermal contact with a large heat bath.

A consequence of Eq. (4.7) is that $\ln Z$ is an extensive quantity that grows proportional with N, the number of particles. By Eqs. (4.4) and (4.5), both $\overline{U}$ and $(\delta U)^2$ are therefore proportional to N which means that the size of *relative fluctuations* in energy behave like

$$\frac{\delta U}{\overline{U}} \propto \frac{\sqrt{N}}{N} = \frac{1}{\sqrt{N}}, \tag{4.8}$$

dacaying to zero in the limit of large systems, where $N \to \infty$ (the so-called *thermodynamic limit*). In particular, for a macroscopic system of $N \approx 10^{24}$ particles, the uncertainty in the energy will be negligibly small. This justifies our previous statement where we equated $\overline{U}$ with the internal energy in thermodynamics. It also means, that any the predictions obtained from the microcanonical ensemble, which has *no* fluctuations in the energy, will become identical to those of the canonical ensemble in this limit. This fact is known as the *equivalence of ensembles.*

4.2 The first law

As we saw, in the statistical description the second law of thermodynamics plays a fundamental role and is closely related to our fundamental assumption. The first law, on the other hand, can be derived from the framework we have developed.

Consider a small change in the system's internal energy $\mathrm{d}U$. This can come either from a change in the possible energy values E_ω available for the system, or from a change in how these energies are occupied, i.e. the probabilities p_ω, or both. We can write this as

$$\mathrm{d}U = \mathrm{d}\left(\sum_\omega p_\omega E_\omega\right) = \sum_\omega p_\omega \mathrm{d}E_\omega + \sum_\omega \mathrm{d}p_\omega E_\omega. \tag{4.9}$$

We will argue that these two terms can be equated with work and heat, respectively. In particular, consider the corresponding change in entropy:

$$\mathrm{d}S = -k_B \sum_\omega \mathrm{d}\left(p_\omega \ln p_\omega\right) = -k_B \sum_\omega \mathrm{d}p_\omega \left(\ln p_\omega + 1\right) = -k_B \sum_\omega \mathrm{d}p_\omega \left(-\frac{E_\omega}{k_B T} - \ln Z + 1\right), \tag{4.10}$$

where in the last equality we used Eq. (4.2) for the probabilities p_ω.

Now, now that however the probabilities p_ω change under the thermodynamic process we are considering, they must maintain their normalization $\sum_\omega p_\omega = 1$. Therefore we have that $\sum_\omega \mathrm{d}p_\omega = \mathrm{d}\left(\sum_\omega p_\omega\right) = 0$ and the last two terms in Eq. (4.10) give zero. Multiplying with T we find that

$$T\mathrm{d}S = \sum_\omega \mathrm{d}p_\omega E_\omega, \tag{4.11}$$

which equals the second term in Eq. (4.9). If we consider a process where the possible energy values E_ω are unchanged then this is the only contribution to $\mathrm{d}U$. In this case, the only change in the state of the system comes from a change in the temperature (which determines p_ω through Eq. (4.2)). The temperature can change due to heat flowing in or out of the system, which is exactly captured by Eq. (4.11).

The first term in Eq. (4.9) on the other hand, corresponds to work. Physically, changes in the energies E_ω come from changing some external parameter in the system, for example by changing the total volume V available for the particles[11], or by tuning some other parameter such as the strength of external electric and magnetic field. Another way to change the energy values is by changing the number of particles N. We thus have that the first term in Eq. (4.9) takes the form

$$\sum_\omega p_\omega \mathrm{d}E_\omega = \underbrace{\left(\sum_\omega p_\omega \frac{\partial E_\omega}{\partial V}\right)}_{\frac{\partial U}{\partial V} = -P} \mathrm{d}V + \underbrace{\left(\sum_\omega p_\omega \frac{\partial E_\omega}{\partial N}\right)}_{\frac{\partial U}{\partial N} = \mu} \mathrm{d}N + \ldots, \tag{4.12}$$

where $\ldots$ corresponds to terms coming from changes in other parameters (e.g. the aforementioned electromagnetic fields). We thus recover the usual thermodynamic expression for the work done on the system.

[11] This is more obvious when we use quantum mechanics to describe the particles. For example, in the problem of a particle confined in a box, the energy eigenvalues depend on the dimensions of the box.

4.3 The third law

Finally, let us consider how the third law of thermodynamics arises in the statistical approach. Take the canonical distribution (4.2) and consider what happens as the temperature T approaches absolute zero. In this limit, the distribution becomes increasingly dominated by states at the lowest possible energies. In particular, when $T \to 0$ the only states that can appear are the ones that correspond to the lowest possible value of E_ω, which we denote by E_0. These are called the system's *ground states* and E_0 is the ground state energy.

Consequently, the Boltzmann entropy (3.3) becomes in this limit

$$\lim_{T\to 0} S = \ln \Omega_0, \tag{4.13}$$

where Ω_0 is the number of distinct ground state configurations that all have the same energy E_0. Now the third law, which states that $\lim_{T\to 0} S = 0$ corresponds to the statement that the ground state is *unique* $\Omega_0 = 1$.

Why should we expect the third law to hold? First, one indeed expects that generically it is unlikely to have multiple states with *exactly* the same energy and thus the ground state is indeed unique. However, one should be a bit more careful. We should remember that we are usually interested in how systems behave in the thermodynamic limit, $N \to \infty$, and this limit might not commute with the limit $T \to 0$. The more physically meaningful way to define the entropy in this limit is to consider a small energy window of size ε and count the number of all states with energies $E_\omega \in [E_0, E_0 + \varepsilon]$ and replace Ω_0 in Eq. (4.13) with this number. Different physical systems will behave differently in terms of how many such low-energy configurations they have. However, what is true of most known physical systems, at least in the absence of some kind of fine-tuning, is that the number of such states grows at most as some power of the number of particles, N^α. In fact, typically one tends to see small powers, such as $\alpha = 0, 1, 2$ etc. This gives a more general formulation of the third law that is indeed satisfied in most known physical systems:

$$\lim_{T\to 0} \frac{S}{N} \propto \frac{\ln(\#\text{ of states with } E_\omega \leq E_0 + \varepsilon)}{N} \propto \frac{\ln N}{N} \to 0 \text{ as } N \to \infty, \tag{4.14}$$

i.e., while the entropy itself might remain finite, the *entropy density* S/N vanishes in the thermodynamic limit.

An interesting and important special case relating to the third law is given by the behavior of *glasses*, such as window glasses (as well as certain magnetic systems with similar properties, known as "spin glasses"). These are experimentally found to violate exhibit some significant *residual entropy* at low temperatures, in violation of the third law. The reason for this is interesting and leads beyond the thermodynamic formalism discussed here. In particular, what happens in glasses is not that they have many states with energies close to the ground state. Instead, they exhibit a large number of so-called *metastable states*. These are configurations whose energies might be much higher than E_0, but ones that correspond to a *local minimum* of the energy in the sense that in order to reach the true ground state the system needs to undergo some complicated rearrangement of many of its constituent particles and has to go through some intermediate states with higher energies in the process. At low temperatures, when the system cannot borrow much energy from its environment, it can therefore get stuck in these metastable configurations for a very long time, indeed for times that are much longer than the time scales on which typical experiments are performed. In this sense, glasses at low temperatures constitute an example where the fundamental assumptions of equilibrium thermodynamics and statistical physics break down, or at least they fail to describe experiments correctly, due to the fact that the system is unable to reach thermal equilibrium. These glasses and glass-like systems constitute an active area of research.